Notes: Chen, Harford, and Lin (JFE, 2015)

Do analysts matter for governance? Evidence from natural experiments

Contents

1 Big picture

- **Research question.** Do financial analysts play a corporate-governance role? More specifically, when a firm loses analyst coverage for reasons unrelated to the firm's own choices, do agency problems become worse?
- **Main idea.** Analysts may monitor managers directly by asking questions, reading disclosures, and scrutinizing decisions. They may also monitor indirectly by spreading information to investors, who can discipline management. The paper tests whether losing analysts weakens this monitoring channel.
- **Identification strategy.** Analyst coverage is endogenous, so the paper uses two natural experiments: brokerage closures and brokerage mergers. These events reduce coverage for some firms because a broker disappears or overlapping analysts are eliminated, not because the covered firm chooses less scrutiny.
- **Empirical design.** The core comparison is from one year before a brokerage exit, $t-1$, to one year after the exit, $t+1$. The paper studies firms that experience exogenous analyst losses and compares them to matched control firms when the outcome requires a difference-in-differences design.
- **Main finding.** After an exogenous decline in analyst coverage, agency problems increase. Shareholders value internal cash less, CEOs receive higher total and excess compensation, acquisitions become worse, and earnings management rises.
- **Where effects are strongest.** The effects are concentrated in firms with low initial analyst coverage and in firms facing less product-market competition. This is important because one lost analyst matters more when few analysts were monitoring the firm in the first place and when product-market discipline is weak.
- **Direct versus indirect monitoring.** The paper finds evidence that analyst losses increase information asymmetry, but institutional ownership does not systematically fall. The governance effects persist even when institutional ownership is non-decreasing, suggesting that analysts themselves are part of the monitoring system rather than only producers of information for institutions.
- **Economic takeaway.** Analysts are not only information intermediaries for pricing securities. They also appear to be external monitors who reduce managerial extraction of private benefits.

2 Incremental contribution of the paper

- **From correlation to causal evidence.** Earlier work often shows that analyst coverage is correlated with firm value, disclosure quality, or governance outcomes. Chen, Harford, and Lin use plausibly exogenous shocks to analyst coverage to move closer to a causal interpretation.
- **Broader governance outcomes.** The paper does not study only forecast accuracy or earnings management. It examines cash policy, CEO compensation, acquisitions, forecast bias, institutional ownership, and earnings management.
- **Agency interpretation of analyst coverage.** The paper directly links analyst losses to outcomes that are classic measures of agency conflicts: the value of cash, excess pay, and

value-destroying acquisitions.

- **Multiple natural experiments.** Broker closures and broker mergers both generate coverage losses. Using both settings increases the number of shocks and reduces concern that one specific event type drives the results.
- **Heterogeneity tests with clear predictions.** The paper predicts stronger effects when the marginal analyst is more important. The evidence fits this prediction: effects are stronger with low initial coverage, less product-market competition, cash-increasing firms, and lower-than-consensus analysts.
- **Monitoring versus information production.** The paper separates the monitoring role of analysts from their information-production role. It shows that the results are not only about market liquidity or institutional investors reacting to less information.
- **Complement to Kelly and Ljungqvist.** Prior evidence shows stock prices fall when analyst coverage disappears. This paper argues that part of the price reaction reflects higher expected agency costs, not only higher information asymmetry.

3 Data and measurement

3.1 Natural experiment sample

The paper constructs exogenous analyst-coverage shocks from brokerage exits between 2000 and 2010.

- **Broker closures.** A broker closure is used when a brokerage disappears from I/B/E/S and the disappearance is confirmed through Factiva or prior broker-closure lists.
- **Broker mergers.** A broker merger is used when two broker houses merge and overlapping analyst coverage creates redundancy. The sample keeps cases in which both houses covered the same firm before the merger and the surviving broker continues to cover the firm.
- **Brokerage exits.** The final sample uses 46 brokerage exits after applying data requirements and sample restrictions.
- **Affected firms.** The main affected-firm sample contains 4,320 firm-year observations for 1,340 unique firms from 1999 to 2011.
- **Event window.** The baseline design keeps only $t-1$ and $t+1$ observations around the brokerage exit. This focuses the analysis on the short-run deviation from the equilibrium level of analyst coverage.

Interpretation. The brokerage-exit design is useful because it changes the supply of analyst monitoring while holding fixed many firm-level demand-side determinants of analyst coverage.

3.2 Analyst coverage and subsamples

- **Coverage.** Analyst coverage is measured using I/B/E/S analyst or broker coverage data.
- **Coverage loss.** In the cash-value sample, the average firm loses about 0.96 analysts after a brokerage exit.

- **Low initial coverage.** A firm is classified as low initial analyst coverage when its pre-exit coverage is in the bottom tercile of the sample. In the cash analysis, the low-coverage subsample has an initial mean of about five analysts.
- **Product-market competition.** More competition means the industry Herfindahl-Hirschman Index is in the bottom quartile of the 48 Fama-French industries. Less competition is the complement.
- **Critical analysts.** A lost analyst is classified as critical if the analyst issued lower-than-consensus recommendations in the year before the brokerage exit.
- **Independent analysts.** Analyst independence follows Gu and Xue's classification of independent versus non-independent brokerage relationships.

Interpretation. The subsamples are designed to test whether the effect of analyst losses is larger when analysts are more likely to be important monitors and when substitute governance mechanisms are weaker.

3.3 Governance outcomes

The paper studies several governance-related outcomes.

- **Marginal value of cash.** The cash analysis asks whether an extra dollar of cash is worth less to shareholders after analyst coverage falls. A lower marginal value of cash is interpreted as investors expecting more waste or diversion of internal funds.
- **CEO compensation.** CEO total compensation is the log of total pay from ExecuComp. CEO excess compensation is the residual from a regression of log total compensation on firm market value and fixed effects.
- **Acquisition decisions.** Acquisition quality is measured using acquirer announcement returns, $CAR(-2, +2)$, and a dummy for whether the CAR is negative.
- **Forecast bias.** Analyst forecast bias is the difference between the forecast EPS and actual EPS, scaled by the previous year's stock price.
- **Institutional ownership.** Institutional ownership is used to test whether analyst losses work mainly by changing institutions' monitoring or information environment.
- **Earnings management.** The paper measures both accrual-based earnings management and real activities manipulation.

3.4 Matched control samples

For CEO compensation, acquisitions, forecast bias, and institutional ownership, the paper uses matched control firms.

- **Portfolio matching.** The strategy follows Derrien and Kecskes, matching by industry, size, Tobin's q, cash flow, and analyst coverage.
- **Propensity-score matching.** The paper also uses nearest-neighbor logit propensity-score matching with relevant firm and deal characteristics.

- **Purpose.** Matching helps compare treated firms to similar firms that did not experience the exogenous analyst loss.

Interpretation. Matching is especially important for outcomes such as CEO compensation and acquisitions, where the paper compares changes in treated firms to changes in similar control firms.

4 Methodology and empirical specifications

4.1 Marginal value of cash holdings

The paper follows Faulkender and Wang's value-of-cash framework. The baseline model is:

$$r_{i,t} - R^B_{i,t} = \alpha_0 + \beta_1 \frac{\Delta Cash_{i,t}}{Mcap_{i,t-1}} + \beta_2 After_i \times \frac{\Delta Cash_{i,t}}{Mcap_{i,t-1}} + \beta_3 After_i + \delta' X + \varepsilon_{i,t}.$$

- $r_{i,t} - R^B_{i,t}$ is the annual excess stock return.
- $\Delta Cash_{i,t}/Mcap_{i,t-1}$ is the change in cash scaled by lagged market capitalization.
- $After_i$ equals one in the post-exit year and zero in the pre-exit year.
- The key coefficient is β_2, the interaction between $After$ and the change in cash.
- A negative β_2 means that the market values an additional dollar of cash less after the exogenous loss of analyst coverage.

Interpretation. The paper interprets a lower post-exit value of cash as evidence that investors expect managers to misuse cash reserves when analyst monitoring weakens.

4.2 CEO compensation difference-in-differences

For compensation, the paper estimates a difference-in-differences effect:

$$DID = \frac{1}{N} \sum_i \Delta CEOComp_i^{Treated} - \frac{1}{N} \sum_i \Delta CEOComp_i^{Control}.$$

- The change is measured from $t - 1$ to $t + 1$ around the brokerage exit.
- The dependent variables are log total CEO compensation and CEO excess compensation.
- Controls enter through matching rather than a single pooled regression specification.
- The paper also examines pay-performance sensitivity before and after the coverage loss for low-coverage firms.

Interpretation. If analyst monitoring restrains private-benefit extraction, CEO pay should rise after an exogenous analyst loss relative to matched controls.

4.3 Acquisition difference-in-differences

For acquisitions, the paper estimates DID effects for announcement returns:

$$DID(CAR) = \frac{1}{N}\sum_i \Delta CAR_i^{Treated} - \frac{1}{N}\sum_i \Delta CAR_i^{Control},$$

and for the probability that the acquirer has a negative announcement return:

$$DID(Negative\ CAR = 1) = \frac{1}{N}\sum_i \Delta NegativeCAR_i^{Treated} - \frac{1}{N}\sum_i \Delta NegativeCAR_i^{Control}.$$

- The acquisition sample uses completed domestic M&A deals by U.S. public acquirers.
- The main return measure is the five-day acquirer CAR, $CAR(-2, +2)$.
- Deal-level variables include relative deal size, tender offers, payment method, private targets, diversifying deals, and high-tech deals.

Interpretation. Worse acquisition announcement returns after analyst losses indicate that managers are more likely to undertake value-destroying acquisitions when external scrutiny falls.

4.4 Forecast bias, analyst characteristics, and channels

The robustness section studies whether analyst coverage affects governance through the characteristics of the lost analysts and through information channels.

- **Forecast bias.** The paper checks whether remaining analysts become more optimistic after a brokerage exit.
- **Critical analysts.** It tests whether effects are stronger when the lost analyst had lower-than-consensus recommendations before exit.
- **Independent analysts.** It tests whether effects differ when the exiting analyst is independent or non-independent.
- **All-star analysts.** The paper studies lost all-star analysts, but small sample sizes limit precision.
- **Information production.** The paper examines illiquidity and institutional ownership to distinguish direct monitoring from indirect information channels.

4.5 Earnings management

The paper revisits analyst coverage and earnings management using the same natural-experiment framework. The baseline model is:

$$Y_{i,t} = \alpha_0 + \beta_1 After + \delta' X + \varepsilon_{i,t},$$

where $Y_{i,t}$ is either the absolute level of accrual-based earnings management or the absolute level of real activities manipulation.

- Accrual-based earnings management is estimated from a modified Jones model.

- Real activities manipulation is based on abnormal production costs and abnormal discretionary expenses.
- The paper uses absolute values because managers can manipulate earnings upward or downward.

Interpretation. If analyst monitoring constrains reporting manipulation, both measures should rise after analyst coverage falls.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies the effect of exogenous decreases in analyst coverage on governance outcomes over a short event window. The evidence is strongest for the claim that analyst monitoring causally restrains some managerial agency problems.

5.2 Why brokerage exits are useful

- Brokerage closures are typically driven by broker-level business conditions rather than covered-firm characteristics.
- Brokerage mergers create analyst redundancy when both merging brokers cover the same firm.
- The paper restricts broker-merger cases to firms covered by both brokers before the merger and still covered by the surviving broker after the merger, reducing concern that firms are dropped endogenously.
- Multiple exits across years allow different firms to be shocked at different times.

Interpretation. The design tries to isolate variation in monitoring supply rather than variation in firms' demand for coverage.

5.3 Main identifying assumptions

- **Exogeneity of brokerage exits.** The exit must affect the firm mainly through the loss of analyst coverage, not through some simultaneous firm-specific shock.
- **Comparable control firms.** Matched controls must represent what would have happened to treated firms absent the analyst loss.
- **Short-run window.** The $t-1$ to $t+1$ window assumes the immediate post-exit period captures the treatment effect before the analyst market fully re-equilibrates.
- **No omitted contemporaneous governance shock.** The results should not be caused by another governance change that happens at the same time as brokerage exits.

Interpretation. The paper addresses these concerns through matching, placebo tests, fixed effects, subsample predictions, and persistence tests.

5.4 Direct monitoring versus information production

Analysts can affect governance in two ways. They can monitor directly by scrutinizing management, or they can produce information that helps investors monitor.

- The paper finds that analyst losses increase stock illiquidity, consistent with an information-production channel.
- Institutional ownership does not significantly change after analyst losses, making it less likely that the main effects are driven only by institutions exiting.
- The cash, compensation, and acquisition results generally hold in both decreasing and non-decreasing institutional-ownership subsamples.

Interpretation. Information production matters, but the evidence gives weight to analysts as direct monitors of managerial behavior.

5.5 Interpretation of the main coefficients

- **Cash value.** In the baseline cash regression, the marginal value of cash falls by about $0.33 after the analyst loss. This is economically large relative to the pre-exit value of cash.
- **CEO pay.** Portfolio-matching results imply that CEO total compensation rises by about 10.8 percent and excess compensation rises by 0.126 after analyst loss.
- **Acquisitions.** Portfolio-matching results imply that acquirer CARs fall by about 1.3 percentage points and the probability of a negative CAR rises by about 9 percentage points.
- **Earnings management.** Accrual-based earnings management rises by 0.036, and real activities manipulation rises by 0.043 after analyst loss.

Economic reading. These coefficients all point in the same direction: less analyst monitoring is followed by more managerial slack.

6 Tables and figures

The paper contains tables rather than plotted figures. The notes below reproduce the tables directly from the paper.

6.1 Tables 1 and 2: Marginal value of cash - baseline evidence

Read Table 1:

- Table 1 reports the sample for the cash-holdings analysis: 3,732 firm-years for 1,179 unique firms from 1999 to 2011.
- Average analyst coverage falls from 12.708 before brokerage exit to 11.748 after brokerage exit.
- The mean change in cash, scaled by lagged market capitalization, is 0.016.

Read Table 2:

- Table 2 estimates the value-of-cash regression.
- The coefficient on $After \times \Delta Cash$ is negative and statistically significant across specifications.
- In the full industry-adjusted specification, the estimate is about -0.333, implying that an additional dollar of cash is worth about \$0.33 less after coverage falls.

Economic message: The market expects internal cash to be used less efficiently after firms lose analyst monitoring.

Table 1
Summary statistics: analysis of the marginal value of cash holdings.
The sample consists of 3,732 firm-years for 1179 unique firms from 1999 to 2011. Variable definitions are given in Appendix B. A * denotes that the variable is scaled by the market value of equity of the firm of fiscal year $t-1$.

	Mean	Standard deviation	Q1	Median	Q3
Analyst coverage					
Before brokerage exit	12.708	7.616	7	12	18
After brokerage exit	11.748	7.481	6	11	17
Excess stock returns during the fiscal year					
Industry-adjusted annual excess stock returns	−0.071	0.510	−0.353	−0.101	0.143
Size and M/B-adjusted annual excess stock returns	−0.020	0.502	−0.325	−0.074	0.171
Firm characteristics					
Total assets (in $ millions)	9,990.350	30,564.900	551.209	2,016.064	7,618.800
Leverage	0.160	0.158	0.025	0.120	0.243
$\Delta Cash_t^*$	0.016	0.327	−0.013	0.004	0.033
$Cash_{t-1}^*$	0.148	0.438	0.025	0.068	0.168
$\Delta Earnings_t^*$	0.134	6.618	−0.020	0.005	0.027
$\Delta NetAssets_t^*$	0.056	0.750	−0.022	0.029	0.103
$\Delta R\&D_t^*$	−0.001	0.041	0	0	0.002
$\Delta Interest_t^*$	0.002	0.048	−0.001	0	0.002
$\Delta Dividends_t^*$	0.000	0.030	0	0	0.001
$NetFinancing_t^*$	0.016	0.250	−0.038	0	0.030

formed based on size and book-to-market ratio; and (2) financial constraint, which is a dummy variable with one

Table 1: Summary statistics for the marginal-value-of-cash analysis.

Table 2
OLS regression analysis of the marginal value of cash holdings.
The sample consists of 3,732 firm-years for 1,179 unique firms from 1999 to 2011. In Columns 1 and 2, the dependent variable is *industry-adjusted excess return* during fiscal year t, and in Columns 3 and 4, it is *Size and M/B-adjusted excess return* of the stock during fiscal year t. *After* is an indicator whose value equals one if the observation is one year after the event of brokerage exit and zero if the observation is one year before brokerage exit. *Constrained* is a dummy variable with one indicating the firm's Whited and Wu (2006) financial constraint index (WW index) is in the top tercile of the sample and zero otherwise. Other variable definitions are given in Appendix B. Financial variables, except *Leverage*, are scaled by the market capitalization of the firm at the end of fiscal year $t-1$. All regressions control for year and Fama and French 48 industry fixed effects, whose coefficient estimates are suppressed. Heteroskedasticity-consistent standard errors clustered at the firm and year levels are reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

	Dependent variable			
	Industry-adjusted annual excess stock returns		Size and M/B-adjusted annual excess stock returns	
	(1)	(2)	(3)	(4)
$\Delta Cash_t$	1.641***	1.801***	1.731***	1.866***
	[0.214]	[0.215]	[0.215]	[0.215]
$After \times \Delta Cash_t$	−0.320**	−0.333**	−0.413***	−0.393***
	[0.134]	[0.135]	[0.134]	[0.135]
$After$	−0.012	−0.011	−0.034	−0.032
	[0.021]	[0.021]	[0.021]	[0.021]
$Cash_{t-1} \times \Delta Cash_t$	−0.074***	−0.179***	−0.076***	−0.207***
	[0.007]	[0.020]	[0.007]	[0.020]
$Leverage_t \times \Delta Cash_t$	−0.587**	−0.174	−0.810***	−0.376
	[0.252]	[0.259]	[0.253]	[0.259]
$Constrained\ (dummy)_t \times \Delta Cash_t$	0.821	1.141*	0.643	0.961*
	[0.583]	[0.582]	[0.585]	[0.582]
$Cash_{t-1}$	0.506***	0.469***	0.473***	0.423***
	[0.039]	[0.040]	[0.039]	[0.040]
$Leverage_t$	−0.503***	−0.455***	−0.632***	−0.589***
	[0.057]	[0.057]	[0.057]	[0.057]
$Constrained\ (dummy)_t$	−0.036**	−0.024	−0.073***	−0.061***
	[0.018]	[0.018]	[0.018]	[0.018]
$\Delta Earnings_t$		0.099***		0.127***
		[0.020]		[0.020]
$\Delta NetAssets_t$		0.143***		0.149***
		[0.020]		[0.020]
$\Delta R\&D_t$		0.215		0.238
		[0.199]		[0.198]
$\Delta Interest_t$		−1.118***		−1.212***
		[0.378]		[0.378]
$\Delta Dividends_t$		0.104		−0.037
		[0.256]		[0.256]
$NetFinancing_t$		−0.190***		−0.161***
		[0.045]		[0.045]
Year fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Number of observations	3,732	3,732	3,732	3,732
Adjusted R^2	0.151	0.171	0.119	0.144

coefficient estimates are suppressed. Across the four mod- (2012), who find that the cumulative abnormal returns

Table 2: Baseline OLS regressions showing that the marginal value of cash falls after analyst coverage decreases.

6.2 Tables 3 and 4: Cash-value heterogeneity and placebo tests

Read Table 3:

- Panel A shows that the cash-value decline is concentrated in firms with low initial analyst coverage and in less competitive industries.
- Panel B shows that the effect appears in cash-increasing firms but not cash-decreasing firms.
- This is consistent with the idea that cash is more vulnerable to agency problems when managers have additional cash to spend.

Read Table 4:

- Table 4 repeats the cash-value analysis using a matched control sample.
- The placebo estimates for matched controls are not statistically significant.
- This weakens the interpretation that the baseline result is only a time trend or mechanical correlation.

Economic message: The cash result appears exactly where the monitoring hypothesis predicts it should be strongest, and it is not reproduced in the matched-control placebo sample.

Table 3
Further exploration of the marginal value of cash holdings: OLS regression analysis.

The sample consists of 3,732 firm-years for 1,179 unique firms from 1999 to 2011. The dependent variable is *Size and M/B-adjusted excess return* of the stock during fiscal year t. In Panel A, the sample is partitioned into low or high analyst coverage before brokerage exit subsamples and less or more product market competition subsamples. A firm is in the Low analyst coverage subsample if the number of analysts following the firm before brokerage exit is in the bottom tercile number of the sample, and in the High analyst coverage subsample if it is in the top tercile of the sample. More competition identifies industries whose HHI index is in the bottom quartile of all 48 Fama and French industries, and Less competition is defined otherwise. HHI is calculated as the sum of squared market shares of all Compustat firms in each industry, and higher values indicate greater concentration and lower product market competitiveness. *Constrained* is a dummy variable with one indicating the firm's Whited and Wu (2006) financial constraint index (WW index) is in the top tercile of the sample, and zero otherwise. In Panel B, the sample is partitioned into cash-increasing and cash-decreasing subsamples. A firm is coded as cash increasing if the cash ratio increases after the exogenous shocks to analyst coverage. *After* is an indicator whose value equals one if the observation is one year after the event of brokerage exit and zero if the observation is one year before brokerage exit. Other variable definitions are given in Appendix B. Financial variables, except *Leverage*, are scaled by the market capitalization of the firm at the end of fiscal year $t-1$. All regressions control for year and Fama and French 48 industry fixed effects, whose coefficient estimates are suppressed. Heteroskedasticity-consistent standard errors clustered at the firm and year levels are reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Analyst coverage before brokerage exits and product market competition

	Dependent variable: *size and M/B-adjusted annual excess stock returns*			
	Low analyst coverage (1)	High analyst coverage (2)	Less competition (3)	More competition (4)
$\Delta Cash_t$	2.135***	2.134***	2.188***	1.656***
	[0.352]	[0.760]	[0.296]	[0.330]
$After \times \Delta Cash_t$	−0.594***	−0.081	−0.684***	−0.300
	[0.208]	[0.362]	[0.197]	[0.195]
$After$	−0.059	−0.029	0.005	−0.062*
	[0.047]	[0.030]	[0.028]	[0.033]
$Cash_{t-1} \times \Delta Cash_t$	−0.163***	−0.790***	−0.186*	−0.177***
	[0.029]	[0.281]	[0.105]	[0.029]
$Leverage_t \times \Delta Cash_t$	0.121	−2.496**	−1.307***	−0.067
	[0.364]	[1.171]	[0.430]	[0.372]
$Constrained$ (dummy)$_t \times \Delta Cash_t$	2.402**	−1.264	0.685	0.859
	[1.047]	[1.875]	[0.751]	[0.960]
$Cash_{t-1}$	0.437***	0.711***	0.493***	0.389***
	[0.069]	[0.106]	[0.059]	[0.058]
$Leverage_t$	−0.570***	−0.740***	−0.631***	−0.564***
	[0.107]	[0.116]	[0.079]	[0.082]
$Constrained$ (dummy)$_t$	−0.051	−0.131***	−0.051**	−0.075***
	[0.038]	[0.040]	[0.024]	[0.027]
$\Delta Earnings_t$	0.097***	0.140	0.186***	0.099***
	[0.029]	[0.090]	[0.028]	[0.030]
$\Delta NetAssets_t$	0.157***	0.110**	0.231***	0.086***
	[0.033]	[0.053]	[0.029]	[0.028]
$\Delta R\&D_t$	0.179	0.336	0.166	−0.063
	[0.267]	[0.761]	[0.212]	[0.551]
$\Delta Interest_t$	−2.876***	2.183	−2.031***	−0.799*
	[0.703]	[1.836]	[0.725]	[0.472]
$\Delta Dividends_t$	−0.830	0.202	−0.119	0.411
	[0.976]	[0.681]	[0.303]	[0.445]
$NetFinancing_t$	−0.121*	−0.394***	−0.360***	0.054
	[0.066]	[0.134]	[0.057]	[0.073]
Year fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Number of observations	1,245	1,252	1,980	1,752
Adjusted R^2	0.158	0.157	0.188	0.145

Table 3, part 1: Cash-value subsamples by initial analyst coverage, product-market competition, and cash increases.

Table 3 (*continued*)

	[1.098]	[0.379]	[1.116]	[0.551]
$Constrained$ (dummy)$_t \times \Delta Cash_t$	2.242	0.688	1.718	0.921**
	[1.542]	[0.643]	[1.859]	[0.456]
$Cash_{t-1}$	0.431***	0.304***	0.419***	0.267***
	[0.121]	[0.082]	[0.109]	[0.086]
$Leverage_t$	−0.449**	−0.269***	−0.621***	−0.365***
	[0.220]	[0.092]	[0.239]	[0.080]
$Constrained$ (dummy)$_t$	−0.043	−0.033	−0.093**	−0.039
	[0.069]	[0.030]	[0.045]	[0.025]
$\Delta Earnings_t$	0.113	0.067	0.159	0.095***
	[0.081]	[0.050]	[0.097]	[0.021]
$\Delta NetAssets_t$	0.178**	0.108***	0.181**	0.113***
	[0.070]	[0.035]	[0.085]	[0.019]
$\Delta R\&D_t$	0.252	0.270	0.066	0.331
	[1.051]	[0.329]	[0.725]	[0.217]
$\Delta Interest_t$	−0.584**	−1.187	−1.126***	−1.150
	[0.280]	[0.725]	[0.385]	[1.275]
$\Delta Dividends_t$	0.087	0.227	−0.249	−0.006
	[0.178]	[0.197]	[0.190]	[0.164]
$NetFinancing_t$	−0.248***	−0.274***	−0.193**	−0.252***
	[0.083]	[0.084]	[0.093]	[0.066]
Year fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Number of observations	2,152	1,532	2,152	1,532
Adjusted R^2	0.233	0.162	0.196	0.130

Table 3, part 2: Continuation of the cash-increasing and cash-decreasing analysis.

Table 4
Analysis of marginal value of cash holdings: placebo tests using the matched control sample.

The table presents placebo tests for the analysis of the marginal value of cash holdings using the matched control sample. The control pool is the remainder of the Compustat universe with valid matching variables. Treated firms are matched using a nearest-neighbor logit propensity score matching. The matching variables are *Size* (*SIZE*), *Leverage* (*LEV*), *Cash* (*CASH*), *Cash flow* (*CF*), and *Coverage* (*COV*) prior to the event of brokerage exit. *Size* is the log of market capitalization of the firm. *Leverage* refers to the leverage ratio, calculated as firm's total debt divided by the market value of total assets. *Cash* is the ratio of cash and short-term investments to total assets. *Cash flow* is calculated as cash flows divided by total assets. *Coverage* refers to the number of analysts covering the firm. The matching also includes industry and year. The sample consists of 3,430 firm-years. Panel A shows the summary statistics for the matched sample, and Panel B reports the placebo test of the marginal value of cash holdings using the matched control sample. In Columns 1 and 2 of Panel B, the dependent variable is *industry-adjusted excess returns* during fiscal year t, and in Columns 3 and 4, it is *Size and M/B-adjusted excess returns* of the stock during fiscal year t. *After* is an indicator whose value equals one if the observation is one year after the event of brokerage exit and zero if the observation is one year before brokerage exit. *Constrained* is a dummy variable with one indicating the firm's Whited and Wu (2006) financial constraint index (WW index) is in the top tercile of the sample and zero otherwise. Other variable definitions are given in Appendix B. Financial variables, except *Leverage*, are scaled by the market capitalization of the firm at the end of fiscal year $t-1$. All regressions control for year and Fama and French 48 industry fixed effects, whose coefficient estimates are suppressed. Heteroskedasticity-consistent standard errors clustered at the firm and year levels are reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Summary statistics for matched samples

Variable	Treated firms			Matched control firms				
	Mean	Standard Deviation	Median	Mean	Standard Deviation	Median	Difference in means	t-Statistic
SIZE	7.655	1.810	7.537	7.403	1.842	7.415	0.252	(0.28)
LEV	0.153	0.152	0.114	0.154	0.155	0.111	−0.001	(−0.10)
CASH	0.172	0.200	0.084	0.174	0.201	0.085	−0.003	(−0.41)
CF	0.076	0.162	0.095	0.079	0.135	0.093	−0.003	(−0.58)
COV	11.688	6.759	11.000	11.743	7.761	10.000	−0.055	(−0.22)

Panel B. Marginal value of cash holdings: placebo tests using the matched control sample

	Dependent variable					
	Industry-adjusted annual excess stock returns		*Size and M/B-adjusted annual excess stock returns*			
					Low analyst coverage	High analyst coverage
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta Cash_t$	1.005**	1.226***	1.085***	1.256***	0.829	2.028*
	[0.419]	[0.463]	[0.408]	[0.474]	[0.691]	[1.199]
$After \times \Delta Cash_t$	−0.241	−0.161	−0.427	−0.352	−0.149	0.055
	[0.259]	[0.243]	[0.279]	[0.261]	[0.444]	[0.384]
After	0.006	0.008	0.010	0.012	0.073	−0.068**
	[0.035]	[0.033]	[0.040]	[0.038]	[0.069]	[0.033]
$Cash_{t-1} \times \Delta Cash_t$	−0.027***	−0.021***	−0.026***	−0.019***	−0.019**	−0.614
	[0.005]	[0.006]	[0.005]	[0.006]	[0.009]	[0.889]
$Leverage_t \times \Delta Cash_t$	−1.019***	−1.203***	−1.015***	−1.253***	−1.378**	−0.381
	[0.274]	[0.395]	[0.238]	[0.377]	[0.576]	[3.229]
Constrained (dummy)$_t \times \Delta Cash_t$	−0.436	0.098	−0.673	−0.358	−1.410	2.479
	[0.882]	[1.006]	[0.942]	[1.058]	[2.131]	[2.689]
$Cash_{t-1}$	0.530***	0.494***	0.473***	0.424***	0.459***	0.628***
	[0.116]	[0.120]	[0.120]	[0.123]	[0.160]	[0.173]
$Leverage_t$	−0.536***	−0.530***	−0.695***	−0.697***	−0.571***	−0.648***
	[0.129]	[0.137]	[0.116]	[0.122]	[0.165]	[0.151]
Constrained (dummy)$_t$	−0.062	−0.056	−0.111*	−0.105*	−0.152***	0.047
	[0.054]	[0.053]	[0.062]	[0.060]	[0.057]	[0.166]
$\Delta Earnings_t$		0.060***		0.064***	0.056*	0.249*
		[0.023]		[0.023]	[0.030]	[0.142]
$\Delta NetAssets_t$		0.090***		0.084*	0.100***	0.027
		[0.034]		[0.043]	[0.038]	[0.050]
$\Delta R\&D_t$		−0.592*		−0.664*	−0.523	3.083**
		[0.356]		[0.384]	[0.338]	[1.244]
$\Delta Interest_t$		0.322		0.358	0.483	−0.023
		[0.374]		[0.386]	[0.430]	[1.855]
$\Delta Dividends_t$		−0.431		−0.564	−1.606*	−0.091
		[0.380]		[0.451]	[0.883]	[0.173]
$NetFinancing_t$		−0.110		−0.084	−0.082*	0.101
		[0.079]		[0.078]	[0.046]	[0.180]
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	3,430	3,430	3,430	3,430	1,137	1,155
Adjusted R^2	0.161	0.178	0.134	0.153	0.196	0.366

Table 4: Placebo tests using matched controls for the marginal value of cash.

Table 5
Summary statistics: analysis of CEO compensation.

The sample consists of 3,516 firm-years from 1999 to 2011. Variable definitions are given in Appendix B.

	Mean	Standard deviation	Q1	Median	Q3
Analyst coverage (COV)					
Before brokerage exit	14.848	7.560	9	14	20
After brokerage exit	13.803	7.350	8	13	18
CEO compensation					
Total compensation (in $ millions)	7.194	7.663	1.975	4.543	9.334
Excess compensation	0.253	0.927	−0.307	0.288	0.858
Log (*Total compensation* in thousands)	8.370	1.075	7.614	8.432	9.154
Firm characteristics					
Size (*SIZE*)	7.992	1.599	6.769	7.847	9.150
Leverage (*LEV*)	0.138	0.136	0.021	0.106	0.209
Tobin's q (*Q*)	2.244	1.654	1.312	1.740	2.600
R&D/Sales (*RDS*)	0.072	0.233	0	0.009	0.095
CapEx/Sales (*CAPXS*)	0.085	0.209	0.025	0.043	0.080
Advertising expense/Sales (*ADVS*)	0.014	0.030	0	0	0.015
Industry-adjusted ROA (*ADJROA*)	0.003	0.096	−0.043	0	0.049
Abnormal stock returns (*ABRET*)	0.134	0.847	−0.227	0.018	0.305
Stock return volatility (*VOL*)	0.395	0.278	0	0	0.553
Institutional ownership (*IO*)	0.734	0.193	0.633	0.762	0.868
Firm age (*AGE*)	24.959	21.239	9	17	35
CEO tenure (*CEOTENURE*)	6.767	6.781	2	5	9

the propensity score match with replacement and keep divide the sample into high versus low analyst coverage

Table 5: Summary statistics for the CEO compensation analysis.

6.3 Tables 5, 6, and 7: CEO compensation and pay-performance sensitivity

Read Table 5:

- Table 5 describes the ExecuComp compensation sample.
- The average CEO total compensation is $7.194 million, and the median is $4.543 million.
- The sample is larger-firm oriented because ExecuComp covers S&P 1500 firms.

Read Table 6:

- Table 6 reports DID estimates for CEO total and excess compensation.
- CEO total compensation rises significantly after analyst loss relative to matched controls.
- The effects are stronger for low initial analyst coverage and less product-market competition.
- The results remain when the same CEO is in place before and after brokerage exit.

Read Table 7:

- Table 7 focuses on low-initial-coverage firms and compares pay-performance sensitivity before and after brokerage exit.
- Before brokerage exit, CEO pay is more sensitive to industry-adjusted ROA and abnormal stock returns.

- After coverage falls, the compensation-performance link weakens.

Economic message: Analyst monitoring appears to restrain CEO pay and helps keep compensation tied to firm performance.

Table 6
Difference-in-differences (DID) analysis of CEO total and excess compensation.

The table presents DID tests for the analysis of CEO compensation using the matched control sample. The control pool is the remainder of the ExecuComp universe with valid matching variables. Treated firms are matched using portfolio matching or nearest-neighbor logit propensity score matching. The matching variables include variants of *Size* (*SIZE*), *Tobin's q* (*Q*), *Cash flow* (*CF*), *Leverage* (*LEV*), *Industry-adjusted ROA* (*ADJROA*), *Abnormal stock returns* (*ABRET*), *Stock return volatility* (*VOL*), *Firm age* (*AGE*), *CEO tenure* (*CEOTENURE*), *R&D/Sales* (*RDS*), *CapEx/Sales* (*CAPXS*), *Advertising expense/Sales* (*ADVS*), *Lagged total compensation* (*LNTDC1*$_{t-1}$), *Insti*$_t$*utional ownership* (*IO*), and *Coverage* (*COV*) prior to the event of brokerage exit. *Size* is the log of market capitalization of the firm. *Tobin's q* is the market value of assets over book value of assets. *Cash flow* is defined as earnings before extraordinary items plus depreciation and amortization divided by total assets. *Leverage* refers to the leverage ratio, calculated as firm's all debt divided by the market value of total assets. *Coverage* refers to the number of analysts covering the firm. The strategy of portfolio matching closely follows Derrien and Kecskes (2013), using *SIZE*, *Q*, *CF*, and *COV*. All matching includes industry and year. The sample consists of 3,516 firm-years for the treatment sample and a same number of firm-years for the control sample. Panel A shows the DID results using portfolio matching as in Derrien and Kecskes (2013), and Panel B reports the DID test of CEO compensation using propensity score matching with variants of matching criteria. In Panel C, we redo the difference-in-differences estimations by dividing the treatment sample into low versus high initial analyst coverage subsamples. In Panel D, we conduct product market competition subsample analysis. Panel E shows the results after dropping 440 cases in which CEO turnover occurs. The dependent variable is either the natural logarithm of *CEO total compensation* or *CEO excess compensation*. *CEO excess compensation* is defined as the residuals from the OLS regression of the natural logarithm of CEO total compensation on the natural logarithm of firms' total market value, industry and year fixed effects in the universal sample of ExecuComp firms. Other variable definitions are given in Appendix B. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

	Mean of DID (treatments versus controls)	
	CEO total compensation	*CEO excess compensation*
Panel A: DID analysis of CEO compensation: portfolio matching on SIZE, Q, CF, and COV		
Mean	0.108**	0.126***
(Standard error)	(0.049)	(0.044)
Panel B: DID analysis of CEO compensation: propensity score matching estimator		
Matching on *SIZE Q, LEV, ADJROA, ABRET, VOL, LNTDC1*$_{t-1}$, *AGE, CEOTENURE, and COV*		
Mean	0.125**	0.135**
(Standard error)	(0.052)	(0.054)
Matching on *SIZE, Q, ADJROA, ABRET, VOL, LNTDC1*$_{t-1}$, *COV*		
Mean	0.106**	0.115**
(Standard error)	(0.051)	(0.053)
Matching on *SIZE, Q, LEV, ADJROA, ABRET, VOL, AGE, CEOTENURE, RDS, CAPXS, ADVS, LNTDC1*$_{t-1}$, and *COV*		
Mean	0.124**	0.134**
(Standard error)	(0.051)	(0.053)
Matching on *SIZE, Q, LEV, ADJROA, ABRET, VOL, AGE, CEOTENURE, RDS, CAPXS, ADVS, LNTDC1*$_{t-1}$, *IO*, and *COV*		
Mean	0.136***	0.145***
(Standard error)	(0.052)	(0.055)
Panel C: DID analysis of CEO compensation: mean difference-in-differences conditional upon analyst coverage before brokerage exits		
Low initial analyst coverage	0.149***	0.138**
(Standard error)	(0.046)	(0.040)
High initial analyst coverage	0.083	0.097
(Standard error)	(0.056)	(0.059)
Panel D: DID analysis of CEO compensation: mean difference-in-differences conditional upon product market competition		
Less competition	0.217***	0.218***
(Standard error)	(0.042)	(0.044)
More competition	0.012	0.032
(Standard error)	(0.041)	(0.042)
Panel E: DID analysis of CEO compensation: same CEO before and after brokerage exits		
Mean	0.111***	0.165***
(Standard error)	(0.047)	(0.051)

Table 6: Difference-in-differences estimates for CEO total and excess compensation.

Table 7
Further exploration of CEO total and excess compensation: changes in pay-performance sensitivity after brokerage exits for firms with low analyst coverage.

In this table, we investigate the changes in pay-performance sensitivities after exogenous brokerage exits for the subset of firms with low initial analyst coverage. The sample consists of 1,078 firm-years from 1999 to 2011. The dependent variable is either the natural logarithm of *CEO total compensation, CEO excess compensation*, or *change in CEO wealth. CEO excess compensation* is defined as the residuals from the OLS regression of the natural logarithm of CEO total compensation on the natural logarithm of firms' total market value, as well as industry and year fixed effects in the universe of ExecuComp firms. Change in CEO wealth is measured by the change in sum of values of unrestricted stock, restricted stock and vested and unvested options held by the CEO. The valuation of options uses a methodology similar to that in Core and Guay (2002). All control variables are lagged measures relative to the year in which compensation is measured. *Institutional ownership* is a dummy variable with one indicating that the total number of shares owned by institutional investors is larger than the annual median and zero otherwise. In Columns 5 and 6, changes in performance measures (*Industry-adjusted ROA* and *Abnormal stock returns*) are used instead of being measured in levels. Other variable definitions are given in Appendix B. All regressions control for Fama and French 48 industry fixed effects, whose coefficient estimates are suppressed. Heteroskedasticity-consistent standard errors clustered at the firm and year levels are reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

	Dependent variable					
	CEO total compensation		*CEO excess compensation*		Δ(*CEO wealth*)	
	Before brokerage exit (1)	After brokerage exit (2)	Before brokerage exit (3)	After brokerage exit (4)	Before brokerage exit (5)	After brokerage exit (6)
Log (*total assets*)	0.530***	0.508***	0.282***	0.282***	−0.070***	−0.048
	[0.032]	[0.032]	[0.041]	[0.027]	[0.019]	[0.078]
Leverage	−0.623	−0.193	−2.269***	−1.936***	0.571*	0.696*
	[0.603]	[0.221]	[0.584]	[0.402]	[0.344]	[0.413]
Tobin's q	0.027	0.022	0.032	0.022	0.087***	0.006
	[0.028]	[0.027]	[0.021]	[0.038]	[0.022]	[0.013]
Industry-adjusted ROA	1.244*	1.217**	0.890	0.672		
	[0.696]	[0.596]	[0.643]	[0.546]		
Abnormal stock returns	0.125***	0.135	0.120***	0.061		
	[0.026]	[0.085]	[0.022]	[0.056]		
Δ*(Industry-adjusted ROA)*					2.325***	−0.427
					[0.731]	[1.237]
Δ*(Abnormal stock returns)*					0.220***	0.050
					[0.030]	[0.049]
Stock return volatility	0.173***	0.143	0.178**	0.375	−0.277**	0.366
	[0.058]	[0.092]	[0.077]	[0.295]	[0.141]	[0.426]
ROA volatility	1.410***	1.291***	1.715***	1.943***	−1.547	−3.771
	[0.234]	[0.489]	[0.324]	[0.698]	[2.240]	[5.096]
R&D/Sales	1.214***	1.528***	1.180*	1.181***	−0.728	−0.378
	[0.436]	[0.425]	[0.609]	[0.349]	[0.767]	[0.984]
CapEx/Sales	−0.122	−0.681***	−0.058	−0.868***	0.323	−0.648
	[0.267]	[0.254]	[0.383]	[0.324]	[0.454]	[0.882]
Advertising expense/Sales	1.238***	0.161	0.060	−1.475	0.353	−1.351
	[0.397]	[0.224]	[0.482]	[1.668]	[4.562]	[1.279]
Firm age	0.002	0.001	−0.001	−0.003	−0.000	0.006***
	[0.001]	[0.001]	[0.002]	[0.002]	[0.006]	[0.002]
CEO tenure	−0.007	−0.001	−0.004	0.001	0.015***	0.010
	[0.004]	[0.005]	[0.005]	[0.006]	[0.004]	[0.007]
Institutional ownership (dummy)	0.188***	0.227***	0.166***	0.220***	0.093	0.033
	[0.044]	[0.044]	[0.027]	[0.070]	[0.115]	[0.227]
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	539	539	539	539	418	418
Adjusted R^2	0.459	0.501	0.286	0.297	0.208	0.213

DID estimates based on both portfolio matching and propensity score matching. The control pool is the remainder of the SDC Mergers and Acquisitions universe with valid matching variables. Treated firms are matched using portfolio matching or a nearest-neighbor logit propensity

Stock dummy (*STOCK*), *All cash* dummy (*ALLCASH*), *Private* dummy (*PRV*), and *Coverage* (*COV*). If multiple deals are conducted by a given acquirer in a particular year, all deal related variables including *CAR* and probability of a negative *CAR* are the mean value of all deals made by that

Table 7: Pay-performance sensitivity before and after brokerage exits for low-coverage firms.

6.4 Tables 8 and 9: Acquisition decisions

Read Table 8:

- Table 8 reports the acquisition sample: 1,464 completed domestic M&A deals.
- The mean five-day acquirer announcement return is slightly negative, and about 51.7 percent of deals have negative CARs.
- The table also reports acquirer and deal characteristics such as size, Tobin's q, relative deal size, stock payment, cash payment, and diversifying deals.

Read Table 9:

- Table 9 shows that acquisition announcement returns decline after exogenous analyst losses.
- Under portfolio matching, CAR falls by about 1.30 percentage points.
- The probability of a negative CAR rises by about 9 percentage points.
- The effects are strongest for firms with low initial analyst coverage and less product-market competition.

Economic message: Analyst monitoring appears to constrain value-destroying empire building through acquisitions.

Table 8
Summary statistics: analysis of acquisition decisions.
The sample consists of 1,464 completed domestic mergers and acquisitions. Variable definitions are given in Appendix B.

	Mean	Standard deviation	Q1	Median	Q3
Analyst coverage					
Before brokerage exit	15.932	9.655	8	15	22
After brokerage exit	15.034	8.673	8	14	21
Acquirer announcement-period abnormal return					
CAR(−2, +2)	−0.228%	6.170%	−3.416%	−0.195%	3.215%
1 if *CAR*(−2, +2) < 0, 0 Otherwise	0.517	0.500	0	1	1
Acquirer characteristics					
Size (*SIZE*)	8.460	1.842	7.148	8.451	9.869
Tobin's q (*Q*)	3.052	4.433	1.413	1.953	3.150
ROA	0.131	0.107	0.086	0.134	0.193
Leverage (*LEV*)	0.113	0.127	0.006	0.071	0.174
Past abnormal returns (*PABRET*)	0.163	1.284	−0.292	0.015	0.321
Total acquisition value in the previous year (*in $ millions*) (ACQ_{t-1})	570.651	1368.151	0	48.500	333
Free cash flows (*FCF*)	0.078	0.111	0.039	0.081	0.139
Deal characteristics					
Relative deal size (*REL*)	0.057	0.120	0.003	0.014	0.054
Friendly deal (*FRD*)	0.979	0.144	1	1	1
Tender offer (*TEND*)	0.040	0.196	0	0	0
Private (*dummy*) (*PRV*)	0.563	0.496	0	1	1
All cash (dummy) (*ALLCASH*)	0.198	0.399	0	0	0
Stock (dummy) (*STOCK*)	0.160	0.366	0	0	0
Diversifying (dummy) (*DIV*)	0.421	0.494	0	0	1
High-tech (dummy) (*HITECH*)	0.461	0.499	0	0	1

Table 8: Summary statistics for acquisition decisions.

Table 9
Difference-in-Differences (DID) analysis of acquisition decisions.
The table presents DID tests for the analysis of acquisition decisions using the matched control sample. The control pool is the remainder of the SDC Mergers and Acquisitions universe with valid matching variables. Treated firms are matched using portfolio matching or a nearest-neighbor logit propensity score matching. The matching variables include variants of *Size* (*SIZE*), *Tobin's q* (*Q*), *Cash flow* (*CF*), *Leverage* (*LEV*), *ROA*, *Past abnormal returns* (*PABRET*), *Free cash flows* (*FCF*), *Total acquisition value in the previous year* (ACQ_{t-1}), *Relative deal size* (*REL*), *Tender offer* (*TEND*), *Friendly deal* (*FRD*), *High-tech dummy* (*HITECH*), *Diversifying dummy* (*DIV*), *Stock dummy* (*STOCK*), *All cash dummy* (*ALLCASH*), *Private dummy* (*PRV*), *and Coverage* (*COV*) prior to the event of brokerage exit. *Size* is the market capitalization of the firm. *Tobin's q* is the market value of assets over book value of assets. *Cash flow* is defined as earnings before extraordinary items plus depreciation and amortization divided by total assets. *Leverage* refers to the leverage ratio, calculated as total debt divided by the market value of total assets. *Coverage* refers to the number of analysts covering the firm. If multiple deals are conducted by a given acquirer in a particular year, all deal related variables are the mean value of all deals made by that particular acquirer in that year, aggregated to the firm level. The strategy of portfolio matching closely follows Derrien and Kecskes (2013), using *SIZE*, *Q*, *CF*, and *COV*. All matching includes industry and year. The sample consists of 2,978 completed domestic mergers and acquisitions made by 651 firms in the treatment sample and a same number of firms in the control sample. Panel A shows the DID results using portfolio matching as in Derrien and Kecskes (2013), and Panel B reports the DID test of acquisitions using propensity score matching with variants of matching criteria. In Panel C, we redo the difference-in-differences estimations by dividing the treatment sample into low versus high initial analyst coverage subsamples. In Panel D, we conduct product market competition subsample analysis. The dependent variable is the acquirer's 5-day cumulative abnormal return (*CAR*) in percentage points, or a dummy variable that equals one if the 5-day *CAR* is negative and zero otherwise. Other variable definitions are given in Appendix B. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

	Mean of DID (treatments versus controls)	
	CAR(−2, +2)	1 if CAR(−2, +2) < 0, 0 Otherwise
Panel A: DID Analysis of acquisition decisions: portfolio matching on SIZE, Q, CF, and COV		
Mean	−1.30%**	9.00%**
(Standard error)	(0.006)	(0.037)
Panel B: DID analysis of acquisition decisions: propensity score matching estimator		
Matching on *SIZE, Q, LEV, ROA, PABRET, FCF,* ACQ_{t-1}, *COV*		
Mean	−0.99%***	5.06%*
(Standard error)	(0.004)	(0.031)
Matching on *SIZE, Q, LEV, ROA, PABRET, FCF,* ACQ_{t-1}, *STOCK, ALLCASH, COV*		
Mean	−1.02%***	5.23%*
(Standard error)	(0.004)	(0.031)
Matching on *SIZE, Q, LEV, ROA, PABRET, FCF,* ACQ_{t-1}, *REL, DIV, TEND, FRD, STOCK, ALLCASH, COV*		
Mean	−1.03%**	5.54%*
(Standard error)	(0.004)	(0.030)
Matching on *SIZE, Q, LEV, ROA, PABRET, FCF,* ACQ_{t-1}, *REL, DIV, TEND, FRD, STOCK, ALLCASH, PRV, COV*		
Mean	−1.06%**	5.67%*
(Standard error)	(0.004)	(0.034)
Panel C: DID analysis of acquisition decisions: mean difference-in-differences conditional upon analyst coverage before brokerage exits		
Low initial analyst coverage	−1.94%***	11.34%**
(Standard error)	(0.007)	(0.046)
High initial analyst coverage	−0.19%	6.14%
(Standard error)	(0.005)	(0.042)
Panel D: DID analysis of acquisition decisions: mean difference-in-differences conditional upon product market competition		
Less competition	−1.95%***	12.24%***
(Standard error)	(0.005)	(0.039)
More competition	−0.60%	4.79%
(Standard error)	(0.005)	(0.036)

Panel D of Table 9. We find the significant effects of analyst coverage on acquisitions exist only for the subset of firms with less market competition. The results indicate that the significant effects on acquisition decisions are mainly evaluating possible channels through which analysts affect governance, alternative model specifications, the effect of star analysts, and the persistence of the effects.

Table 9: Difference-in-differences analysis of acquisition announcement returns and negative-CAR probabilities.

6.5 Tables 10, 11, and 12: Robustness, analyst characteristics, and channels

Read Table 10:

- Table 10 studies analyst forecast bias.
- The mean and median optimistic forecast bias rise after an exogenous loss of coverage.
- This suggests that remaining analysts become more optimistic when analyst competition or scrutiny falls.

Read Table 11:

- Table 11 asks whether the identity of the lost analyst matters.
- Effects are stronger when the exiting analyst was lower-than-consensus, meaning the lost analyst was more critical.
- The table also compares independent and non-independent analysts.

Read Table 12:

- Table 12 investigates whether institutional ownership changes after analyst losses.
- The DID estimate for institutional ownership is not significant.

- The CEO compensation and acquisition effects generally hold in both decreasing and non-decreasing institutional-ownership subsamples.
- The cash-value regressions are also informative: the cash result is significant even when institutional ownership does not decrease.

Economic message: The robustness evidence points toward a direct governance role for analysts, not merely a mechanical information-production effect through institutional investors.

Table 10

Difference-in-differences (DID) analysis of analyst forecast bias.

The table presents DID tests for the analysis of analyst forecast bias using the matched control sample. Treated firms are matched using nearest-neighbor logit propensity score matching. The matching variables are *Size* (*SIZE*), *Tobin's q* (*Q*), *Leverage* (*LEV*), *Cash* (*CASH*) *and Coverage* (*COV*). *Size* is the log of market capitalization of the firm. *Tobin's q* is the market value of assets over book value of assets. *Leverage* refers to the leverage ratio, calculated as total debt divided by the market value of total assets. *Cash* is the ratio of cash and short-term investments to total assets. *Coverage* refers to the number of analysts covering the firm. The main variables of interest are *Mean bias* and *Median bias*. *Analyst forecast bias* is the difference between the forecast analyst j in year t and the actual EPS, expressed as a percentage of the previous year's stock price. *Consensus bias* is expressed as a mean or median bias among all analysts covering a particular stock. We exclude observations that fall below the 25th percentile of the size distribution, observations with stock prices lower than $5, and those for which the absolute difference between forecast value and the true earnings exceeds $10. The sample consists of 3292 firm-years for the treatment sample and a same number of firm-years for the control sample. In Panel A, summary statistics are presented. Panel B provides DID estimator results. Other variable definitions are given in Appendix B. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Summary statistics for the treatment sample

	Mean	Standard deviation
Mean bias	0.0381	0.1110
Median bias	0.0380	0.1180

Panel B: DID results

	Mean of DID (treatments versus controls)	
	Mean bias	*Median bias*
Matching on *SIZE, Q, LEV, CASH,* and *COV*		
Mean of DID	0.0093**	0.0086**
(Standard error)	(0.0037)	(0.0039)

Table 10: Difference-in-differences analysis of analyst forecast bias.

Table 11

Difference-in-differences (DID) analysis of CEO compensation and acquisition decisions: exiting analyst was lower than consensus or independent.

The table presents DID tests for the analysis of CEO compensation acquisition decisions using the matched control sample, conditional upon whether or not the exiting analyst issued a lower investment recommendation about the affected stock than consensus or was independent in the year before the brokerage exit. The classification of analyst independence follows Gu and Xue (2008). Treated firms are matched using nearest-neighbor logit propensity score matching. The matching variables follow Tables 6 and 9. The sample consists of 605 firm-years in the lower-than-consensus subsample and 2,907 firm-years in the others subsample in Panel A and 220 firm-years in the lower-than-consensus subsample and 924 firm-years in the others subsample in Panel B. Others include higher-than-consensus and equal-to-consensus observations. The sample consists of 432 firm-years in the independent analyst subsample and 3,082 firm-years in the non-independent analyst subsample in Panel C and 170 firm-years in the independent analyst subsample and 1,162 firm-years in the non-independent analyst subsample in Panel D. In Panels A and C, the dependent variable is either the natural logarithm of *CEO total compensation* or *CEO excess compensation*. *CEO excess compensation* is defined as the residuals from the OLS regression of the natural logarithm of *CEO total compensation* on the natural logarithm of the firm's total market value, as well as industry and year fixed effects in the sample of ExecuComp firms. The dependent variable in Panels B and D is the acquirer's 5-day cumulative abnormal return (*CAR*) in percentage points, or a dummy variable which equals one if the 5-day *CAR* is negative and zero otherwise. Other variable definitions are given in Appendix B. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: DID analysis of CEO compensation: lower-than-consensus versus others

	Mean of DID (treatments versus controls)	
	CEO total compensation	*CEO excess compensation*
Lower-than-consensus	0.202***	0.212***
(Standard error)	(0.071)	(0.072)
Others	0.106***	0.114***
(Standard error)	(0.032)	(0.034)

Panel B: DID analysis of acquisition decisions: lower-than-consensus versus others

	Mean of DID (treatments versus controls)	
	$CAR(-2, +2)$	1 if $CAR(-2, +2) < 0$, 0 Otherwise
Lower-than-consensus	−1.86%*	13.78%**
(Standard error)	(0.010)	(0.06)
Others	−0.91%**	2.54%
(Standard error)	(0.005)	(0.029)

Panel C: DID analysis of CEO compensation: independent versus non-independent

	Mean of DID (treatments versus controls)	
	CEO total compensation	*CEO excess compensation*
Independent	0.150*	0.152*
(Standard error)	(0.087)	(0.088)
Non-independent	0.119***	0.129***
(Standard error)	(0.031)	(0.033)

Panel D: DID analysis of acquisition decisions: independent versus non-independent

	Mean of DID (treatments versus controls)	
	$CAR(-2, +2)$	1 if $CAR(-2, +2) < 0$, 0 Otherwise
Independent	−1.86%**	13.82%**
(Standard error)	(0.001)	(0.063)
Non-independent	−0.95%***	4.48%*
(Standard error)	(0.003)	(0.025)

the firm in the three years before its exit. The underwriting which polls buy-side institutional investors every year to

Table 11: Effects when the exiting analyst was lower-than-consensus or independent.

Table 12

Analysis of CEO compensation, acquisition decisions, and the marginal value of cash holdings: decreasing versus non-decreasing institutional ownership.

The table reports the analysis of CEO compensation, acquisition decisions, and the marginal value of cash holdings, conditional upon whether institutional ownership is decreasing or non-decreasing following the events of broker exit. Panel A reports summary statistics for institutional ownership in the treatment sample prior to brokerage exits. Panel B reports the DID results of institutional ownership. Panels C and D report the DID results of CEO compensation and acquisition decisions, conditional upon whether institutional ownership is decreasing or non-decreasing following brokerage exits. Panel E reports the OLS regression results of the marginal value of cash, according to whether institutional ownership or liquidity is increasing or decreasing following the events of broker exit. The sample ranges from 1999 to 2011. Treated firms are matched using nearest-neighbor logit propensity score matching. The matching variables in Panel B include *SIZE, Q, LEV, CASH, IO*, and *COV*. The matching variables in Panels C and D follow Tables 6 and 9, respectively. The sample contains 1,162 firm-years in the *IO* decreasing subsample in Panel C and 664 acquisition deals in the *IO* decreasing subsample in Panel D. In Panel C, the dependent variable is either the natural logarithm of *CEO total compensation* or *CEO excess compensation. CEO excess compensation* is defined as the residuals from the OLS regression of the natural logarithm of *CEO total compensation* on the natural logarithm of the firm's total market value, as well as industry and year fixed effects in the sample of ExecuComp firms. The dependent variable in Panel D is the acquirer's 5-day cumulative abnormal return (*CAR*) in percentage points or a dummy variable that equals one if the 5-day *CAR* is negative and zero otherwise. In Panel E, the dependent variable is *Size and M/B-adjusted excess returns* of the stock during fiscal year *t*. *After* is an indicator whose value equals one if the observation is one year after the event of brokerage exit and zero if the observation is one year before brokerage exit. *Constrained* is a dummy variable with one indicating the firm's Whited and Wu (2006) financial constraint index (WW index) is in the top tercile of the sample and zero otherwise. Other variable definitions are given in Appendix B. Financial variables, except *Leverage*, are scaled by the market capitalization of the firm at the end of fiscal year $t-1$. All regressions control for year and Fama and French 48 industry fixed effects, whose coefficient estimates are suppressed. Heteroskedasticity-consistent standard errors clustered at the firm level are reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Summary statistics for the treatment sample

	Mean	Std. dev.
Full sample	0.6174	0.2814
Low analyst coverage	0.4753	0.2940
High analyst coverage	0.6852	0.2331

Panel B: DID results of institutional ownership

	Mean of DID (treatments versus controls) *Institutional Ownership*
Matching on *SIZE, Q, LEV, CASH, IO*, and *COV*	
Full sample	−0.0028
(Standard error)	(0.0070)
Low analyst coverage	0.0107
(Standard error)	(0.0170)
High analyst coverage	−0.0059
(Standard error)	(0.0087)

Panel C: DID analysis of CEO compensation: decreasing versus non-decreasing institutional ownership

	Mean of DID (treatments versus controls)	
	CEO total compensation	*CEO excess compensation*
Decreasing	0.113**	0.131**
(Standard error)	(0.054)	(0.058)
Non-decreasing	0.139***	0.146***
(Standard error)	(0.035)	(0.036)

Panel D: DID analysis of acquisition decisions: decreasing versus non-decreasing institutional ownership

	Mean of DID (treatments versus controls)	
	$CAR(-2, +2)$	1 if $CAR(-2, +2) < 0$, 0 Otherwise
Decreasing	−1.37%**	6.22%*
(Standard error)	(0.004)	(0.035)
Non-decreasing	−0.63%*	4.58%*
(Standard error)	(0.004)	(0.025)

Panel E: Decreasing versus non-decreasing institutional ownership

	Dependent variable			
	Industry-adjusted annual excess stock returns		*Size and M/B-adjusted annual excess stock returns*	
	Decreasing	Non-decreasing	Decreasing	Non-decreasing
	(1)	(2)	(3)	(4)
$\Delta Cash_t$	1.081***	2.742***	1.134**	2.709***
	[0.172]	[0.349]	[0.475]	[0.210]
$Afte \times \Delta Cash_t$	−0.259	−0.540***	−0.494***	−0.479**
	[0.186]	[0.186]	[0.089]	[0.229]

Table 12, part 1: Institutional ownership, CEO compensation, acquisitions, and cash-value tests.

Table 12 (*continued*)

After	0.023	−0.027	0.014	−0.056**
	[0.027]	[0.022]	[0.035]	[0.026]
$Cash_{t-1} \times \Delta Cash_t$	0.004	−0.140***	0.159	−0.161***
	[0.125]	[0.033]	[0.126]	[0.046]
$Leverage_t \times \Delta Cash_t$	−0.189	−0.302	−0.752	−0.541
	[0.653]	[0.388]	[0.888]	[0.424]
Constrained $(\text{dummy})_t \times \Delta Cash_t$	−0.919	3.337***	−1.285	3.150***
	[2.213]	[0.448]	[2.996]	[0.335]
$Cash_{t-1}$	0.463***	0.517***	0.498***	0.440***
	[0.090]	[0.174]	[0.097]	[0.162]
$Leverage_t$	−0.428***	−0.348**	−0.577***	−0.441***
	[0.163]	[0.136]	[0.160]	[0.150]
Constrained $(\text{dummy})_t$	−0.130***	0.088	−0.197***	0.080
	[0.049]	[0.057]	[0.025]	[0.057]
$\Delta Earnings_t$	0.149***	0.070***	0.205***	0.098***
	[0.038]	[0.024]	[0.061]	[0.034]
$\Delta NetAssets_t$	0.097**	0.202***	0.085	0.212***
	[0.044]	[0.035]	[0.067]	[0.034]
$\Delta R\&D_t$	−1.416***	0.407***	−1.133***	0.517***
	[0.259]	[0.099]	[0.224]	[0.160]
$\Delta Interest_t$	0.596*	−3.512***	−0.122	−3.844***
	[0.345]	[0.657]	[0.443]	[1.007]
$\Delta Dividends_t$	−0.222	0.644*	−0.365***	−0.305
	[0.173]	[0.372]	[0.141]	[0.204]
$NetFinancing_t$	−0.263**	−0.237	−0.108	−0.227
	[0.124]	[0.146]	[0.159]	[0.141]
Year fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Number of observations	1,630	1,702	1,630	1,702

Table 12, part 2: Continuation of cash-value regressions by institutional-ownership changes.

6.6 Appendix tables: Exits, variable definitions, and earnings management

Read Appendix Table A1:

- The appendix reports the distribution of brokerage exits across years.
- The sample contains 46 brokerage exits and 2,160 affected firm observations at the firm-year level used in the appendix count.
- Exits are spread across 2000-2010 rather than concentrated in one year.

Read Appendix Table B1:

- Table B1 defines the variables used across the cash, compensation, acquisition, earnings-management, and institutional-ownership analyses.
- It also lists data sources such as I/B/E/S, Factiva, SDC, Compustat, CRSP, ExecuComp, and CDA/Spectrum 13(f) filings.

Read Appendix Table C1:

- Table C1 revisits analyst coverage and earnings management.
- The coefficient on $After$ is positive and significant for both accrual-based earnings management and real activities manipulation.

- This supports the view that analyst monitoring also constrains financial reporting manipulation.

Economic message: The appendices document the treatment events, define the empirical variables, and show that the analyst-monitoring result extends to earnings management.

Table A1
Descriptive statistics for brokerage exits.
The table presents the number of brokerage exits and the number of affected firms in each year from 2000 to 2010 in our sample.

Year	Number of brokerage exits	Number of affected firms
2000	7	364
2001	9	388
2002	4	316
2003	2	26
2004	2	76
2005	7	200
2006	3	103
2007	5	309
2008	3	266
2009	3	28
2010	1	84
Total	46	2,160

Appendix Table A1: Brokerage exits and affected firms by year.

Variable definitions.

Variable	Definition	Source
After	Indicator equal to one if the observation is one year after the event of brokerage exit and zero if the observation is one year before brokerage exit.	Institutional Brokers' Estimate System (I/B/E/S), Factiva, Thomson Reuters Securities Data Company (SDC), and Kelly and Ljungqvist (2012)
Analysis of the marginal value of cash holdings		
Industry-adjusted annual excess stock returns	Firm-level stock returns minus Fama and French (1997) industry value-weighted returns.	Ken French's website: http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
Size and M/B adjusted annual excess stock returns	Firm-level stock returns minus Fama–French size and market-to-book matched portfolio returns.	Ken French's website: http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
Leverage	Total debt (*dltt*+*dlc*)/Market value of total assets (*at*−*ceq*+*csho*×*prcc_f*).	Compustat
Constrained (dummy)	Dummy variable with one indicating that the firm's Whited and Wu (2006) financial constraint index (WW index) is in the top tercile of the sample and zero otherwise.	Compustat
$\Delta Cash_t$	Change in cash (*che*).	Compustat
$\Delta Earnings_t$	Change in earnings before extraordinary items (*ib*+*xint*+*txdi*+*itci*).	Compustat
$\Delta NetAssets_t$	Change in net assets (*at*−*che*).	Compustat
$\Delta R\&D_t$	Change in research and development expense (*xrd*, set to zero if missing).	Compustat
$\Delta Interest_t$	Change in interest (*xint*).	Compustat
$\Delta Dividends_t$	Change in common dividends (*dvc*).	Compustat
$NetFinancing_t$	New equity issues (*sstk*−*prstkc*)+Net new debt issues (*dltis*−*dltr*). Source: Compustat	Compustat
Analysis of CEO compensation		
Total compensation (in $ millions)	Annual total compensation received by a Chief Executive Officer (CEO), comprising salary, bonus, restricted stock awards, stock option grants, long-term incentive payouts, and all others.	ExecuComp
Excess compensation	Residuals from the ordinary least squares regression of the natural logarithm of CEO total compensation on the natural logarithm of firms' total market value, as well as industry fixed effects and year fixed effects in the sample of ExecuComp firms.	ExecuComp
Size (*SIZE*)	Log of the market capitalization of the firm (*prcc_f*×*csho*).	Compustat
Leverage (*LEV*)	Total debt (*dltt*+*dlc*) / Market value of total assets (*at*−*ceq*+*csho*×*prcc_f*).	Compustat
Tobin's q (*Q*)	Market value of assets over book value of assets: (*at*−*ceq*+*csho*×*prcc_f*)/*at*.	Compustat
R&D/Sales (*RDS*)	*xrd* (set to zero if missing)/*sale*.	Compustat
CapEx/Sales (*CAPXS*)	*capx* (set to zero if missing)/*sale*.	Compustat
	xad (set to zero if missing)/*sale*.	Compustat

Appendix Table B1, part 1: Variable definitions and sources.

Table B1 (*continued*)

Variable	Definition	Source
Advertising expense/Sales (*ADVS*)		
Industry-adjusted ROA (*ADJROA*)	Return on assets (ROA) (*oibdp*/*at*) adjusted by industry median ROA. Industries are defined according to Fama and French 48 industry classifications.	Ken French's website: http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
Abnormal stock returns (*ABRET*)	Compounded daily excess (over the Center for Research in Security Prices (CRSP) value-weighted index) returns over the fiscal year.	CRSP
Stock return volatility (*VOL*)	Standard deviation of monthly stock returns during the past five years.	ExecuComp and CRSP
Firm age (*AGE*)	Number of years since a firm's first appearance in CRSP.	CRSP
CEO tenure (*CEOTENURE*)	Number of years a CEO has been in office.	ExecuComp
Analysis of acquisition decisions		
CAR(−2, +2)	Five-day cumulative abnormal return calculated using a market model estimated over the period [−210, −11] relative to the announcement date (day 0).	CRSP
Size (*SIZE*)	Log of the market capitalization of the firm (*prcc_f*×*csho*).	Compustat
Tobin's q (*Q*)	Market value of assets over book value of assets: (*at*−*ceq*+*csho* ×*prcc_f*)/*at*.	Compustat
ROA	Operating income before depreciation (*oibdp*) over book value of total assets (*at*).	Compustat
Leverage (*LEV*)	Total debt (*dltt*+*dlc*)/Market value of total assets (*at*−*ceq*+*csho*×*prcc_f*).	Compustat
Past abnormal returns (*PABRET*)	Compounded daily excess (over the CRSP value-weighted index) returns over prior fiscal year.	CRSP
Total acquisition value in the previous year (ACQ_{t-1})	Summation of the deal value of all the transactions made by a given acquirer in the previous year.	SDC
Free cash flows (*FCF*)	(*oibdp- xint-txdi-capx*)/*at*.	Compustat
Relative deal size (*REL*)	Deal value (from SDC) over acquirer's market value of total assets (*at* −*ceq*+*csho*×*prcc_f*).	SDC and Compustat
Friendly deal (*FRD*)	Indicator variable equals one for friendly deals and zero otherwise.	SDC
Private (dummy) (*PRV*)	Indicator variable equals one for private targets and zero otherwise.	SDC
All cash (dummy) (*ALLCASH*)	Indicator variable equals one for purely cash-financed deals and zero otherwise.	SDC
Stock (dummy) (*STOCK*)	Indicator variable equals one for deals at least partially stock financed and zero otherwise.	SDC
Diversifying (dummy) (*DIV*)	Indicator variable equals one if acquirer and target do not share a Fama and French industry and zero otherwise.	SDC
Tender offer (*TEND*)	Indicator variable equals one if the deal involves a tender offer and zero otherwise.	SDC
High-tech (dummy) (*HITECH*)	Indicator variable equals one if acquirer and target are both from the high-tech industries defined by Loughran and Ritter (2004) and zero otherwise	SDC and Loughran and Ritter (2004)
Analysis of earnings management		
Absolute level of accrual-based earnings management	Absolute value of discretionary accruals, estimated by a cross-sectional version of the modified Jones (1991) model	Compustat
Absolute level of real activities manipulation	Calculated using abnormal level of production costs and abnormal level of discretionary expenditures, estimated cross-sectionally following Roychowdhury (2006) and Zang (2012)	Compustat
Market-to-book ratio	(Fiscal-year-end market value of equity+book value of liabilities)/total assets.	Compustat
ROA	Operating income before depreciation (*oibdp*) over book value of total assets (*at*).	Compustat
Growth rate of assets	Change of assets (*at*) scaled by lagged assets.	Compustat
Cash flow volatility	Standard deviation of cash flows of a firm in the entire sample period, scaled by lagged assets. Source: Compustat	Compustat
External financing activities	Measured by the sum of net cash received from equity and debt issuance scaled by total assets. Source: Compustat	Compustat
Size	Log of the market capitalization of the firm (*prcc_f*×*csho*).	Compustat
Institutional ownership	Institutional ownership is measured by the percentage of common shares owned by institutional investors.	CDA/Spectrum Institutional 13(f) filings

Appendix Table B1, part 2: Continued variable definitions and sources.

Table C1
Revisiting the effect of analyst coverage on earnings management.
The sample consists of 4,320 firm-years for 1,340 unique firms from 1999 to 2011. Panel A provides summary statistics, while Panel B presents OLS regression results. Variable definitions are given in Appendix B. The dependent variable in Column 1 is the absolute value of discretionary accruals, estimated by a cross-sectional version of the modified Jones (1991) model. In Column 2, the dependent variable is the absolute level of real activities manipulation, calculated using abnormal level of production costs and abnormal level of discretionary expenditures, estimated cross-sectionally following Roychowdhury (2006) and Zang (2012). *After* is an indicator whose value equals one if the observation is one year after the event of brokerage exit and zero if the observation is one year before brokerage exit. *Institutional ownership* is measured by the percentage of common shares owned by institutional investors. Other variable definitions are given in Appendix B. All regressions control for year and industry fixed effects, whose coefficient estimates are suppressed. Heteroskedasticity-consistent standard errors clustered at the firm level are reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A. Summary statistics

	Mean	Standard deviation	Q1	Median	Q3
Analyst coverage					
Before brokerage exit	12.581	7.714	6	11	18
After brokerage exit	11.748	7.697	6	11	17
Earnings management					
Absolute level of accrual-based earnings management	0.192	0.464	0.024	0.059	0.148
Absolute level of real activities manipulation	0.382	0.593	0.103	0.248	0.486
Firm characteristics					
Market-to-book ratio	2.193	1.758	1.253	1.697	2.546
ROA	0.115	0.148	0.076	0.130	0.187
Growth rate of assets	0.150	0.425	−0.024	0.068	0.191
Cash flow volatility	0.117	0.184	0.000	0.069	0.118
External financing activities	0.017	0.253	−0.036	−0.001	0.026
Size	8.046	1.871	6.731	7.978	9.335
Institutional ownership	0.631	0.283	0.479	0.703	0.842

Panel B: OLS regression analysis of earnings management

	Dependent variable	
	Absolute level of accrual-based earnings management (1)	*Absolute level of real activities manipulation* (2)
After	0.036***	0.043***
	[0.014]	[0.014]
ROA	−0.136**	−0.153**
	[0.061]	[0.074]
Growth rate of assets	0.113***	0.232***
	[0.027]	[0.053]
Cash flow volatility	0.136**	0.300*
	[0.063]	[0.157]
External financing activities	−0.025	−0.034
	[0.067]	[0.110]
Size	−0.002	−0.027***
	[0.004]	[0.007]
Market-to-book ratio	−0.017***	0.048***
	[0.006]	[0.008]
Institutional ownership	−0.001	0.011
	[0.024]	[0.034]
Year fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
Number of observations	4,320	3,806
Adjusted R^2	0.185	0.200

Appendix C

Bhushan, R., 1989. Firm characteristics and analyst following. Journal of Accounting and Economics 121, 255–274.
Bushman, R., Piotroski, J., Smith, A., 2005. Insider trading restrictions and

Appendix Table C1: Analyst coverage and earnings management.

7 Short summary

- The paper studies whether analysts matter for corporate governance.
- Analyst coverage is likely endogenous, so the paper uses brokerage closures and mergers as natural experiments that reduce coverage exogenously.
- The sample contains 46 brokerage exits from 2000 to 2010 and affected firms observed around the exit.
- The first major outcome is the marginal value of cash. After analyst coverage falls, shareholders value additional cash less.
- The cash-value result is strongest for low-coverage firms, less competitive industries, and cash-increasing firms.
- The second outcome is CEO compensation. After coverage falls, CEO total and excess compensation rise relative to matched controls.
- The CEO pay effect is strongest when initial analyst coverage is low and product-market competition is weak.
- Pay-performance sensitivity also weakens after the exogenous loss of analyst coverage.
- The third outcome is acquisition quality. After coverage falls, acquirer announcement returns decline and negative-CAR acquisitions become more likely.
- Remaining analysts become more optimistic after coverage losses, and effects are stronger when the lost analyst was more critical.
- Institutional ownership does not significantly fall after analyst losses, suggesting that results are not driven only by institutional investors exiting.
- Earnings management rises after analyst coverage falls.
- Overall, analysts appear to serve as external monitors who reduce agency costs and protect outside shareholders.

8 Conclusion

8.1 Core conclusion

Chen, Harford, and Lin show that analysts matter for corporate governance. When firms lose analyst coverage because of brokerage closures or mergers, several governance outcomes deteriorate. Cash is worth less to shareholders, CEOs receive higher pay, acquisitions become worse, and managers engage in more earnings management.

8.2 Economic intuition

Analysts impose scrutiny. They follow financial statements, ask questions, issue reports, and communicate information to investors. This scrutiny makes it harder for managers to waste cash, extract excessive pay, make empire-building acquisitions, or manipulate earnings. When an analyst disappears, the cost of managerial opportunism falls.

8.3 Why the heterogeneity matters

The strongest results occur where the marginal analyst should matter most: firms with few analysts, firms in less competitive industries, and firms losing critical analysts. These patterns are difficult to explain with a simple omitted-variable story and are consistent with the monitoring hypothesis.

8.4 Interpretation relative to information asymmetry

The paper does not deny that analysts produce information. Stock liquidity worsens after analyst loss, which is consistent with an information channel. But the evidence does not support an interpretation based only on institutional investors losing information. Institutional ownership does not significantly decline, and the governance effects persist in non-decreasing institutional-ownership subsamples.

8.5 Implications

The paper implies that the analyst industry has real governance consequences. A decline in coverage is not only a capital-market information event; it can change managerial behavior. This matters for firms with low coverage, for small or opaque firms, and for periods when brokerage capacity contracts.

8.6 Limitations

The evidence is based on natural experiments and matched comparisons, not randomized assignment of analyst coverage. Brokerage exits may still affect firms through multiple channels, including liquidity and information asymmetry. Some subsample tests have limited power, especially all-star analyst tests. The paper is strongest on the short-run effect of coverage losses around brokerage exits.

8.7 Takeaway

The enduring takeaway is that financial analysts are part of the governance ecosystem. Their coverage disciplines managers, increases the value of internal cash, restrains CEO pay, improves acquisition quality, and reduces earnings management.